
ASO Operations, Testing, ALM, and Release Gates

Observability, dashboards, alerts, E2E tests, ALM, customer-ready checklist, and approval gates.

GitHub path: docs/architecture/ASO_Operations_Testing_ALM_and_Release_Gates.md

Version 1.1 - public customer-ready split package - refined for readability and GitHub publishing

Safety note: this public reference pack must not contain tenant IDs, credentials, secrets, real customer data, private endpoints, or connection strings.

Operations, testing, ALM, and release gates

This document centralizes the operational standards that are distributed throughout the end-to-end guide: observability, tests, ALM, release evidence, runbooks, and approval gates.

End-to-end testing before HubSpot

Test	Expected result	Pass criteria
New lead, no SAP match	Lead simulated in Dataverse; Foundry qualifies; SAP search no match; Dataverse updated; Lead Nurture starts if eligible.	No duplicate lead; Agent Run Log exists; AI fields populated; Communication State correct; Journey ledger row exists.
Existing SAP customer lead	Lead matches SAP; Foundry calls SAP wrapper via APIM; existing customer review flagged if needed; seller task created.	No direct ERP write; SAP context normalized; review flag set if ambiguous.
Opportunity risk and deal guidance	Opportunity agent generates risk/recommendation; Foundry validates; Power Automate updates fields and creates task.	Seller sees guidance; no unauthorized communication or ERP action.
Commercial approval and SAP submit	Pending action created; approval required; APIM called after approval; SAP reference returned; Dataverse updated.	Approval captured; idempotency key present; SAP reference persisted; no blind retry.
Sales agent and Foundry disagree	Sales Qualification Agent says strong fit but SAP/Foundry says ambiguous; Foundry requires review.	No automatic AE routing; review task created; disagreement logged.

Operations dashboards

- Runs by status.
- Failed runs by component.
- Average confidence by agent type.
- Leads requiring human review.
- Opportunities high risk.
- Journey failures.
- SAP submit failures.
- Pending commercial approvals.

Alerts

- SAP wrapper 5xx spike.
- APIM unauthorized calls.
- Foundry failure rate above threshold.
- Invalid JSON/schema failures.

- Flow retry storm.
- Customer Insights send volume drop.
- Journey start failures.
- Unsubscribe/bounce anomaly.
- Pending commercial actions stuck beyond SLA.

ALM and release standard

Stage	Standard
Build	Build unmanaged only in development/trial build environment, inside approved ASO solutions.
Export	Export manual backups after major milestones in trial; use pipelines for enterprise target.
Promotion	Enterprise target uses managed solutions from DEV through SIT/UAT/PREPROD/PROD.
Versioning	Trial build starts at 0.1.0.0; release candidates use 1.0.0-rc where applicable; production uses semantic versioning.
Source control	Use Git integration, SolutionPackager, Power Platform CLI, or pipelines where available in enterprise delivery.
Public GitHub	Publish only sanitized architecture, standards, sample schemas, mock data, and public-safe screenshots.

Customer-ready implementation checklist

Checkpoint	Required evidence
Architecture boundaries approved	HubSpot ingress only, Dataverse operational truth, Customer Insights only sends, Foundry orchestrates, Power Automate executes, SAP via APIM/Functions.
Naming standard approved	Schema names, variables, choices, flows, views, Azure resources, journeys, agents, and APIs use the standard.
Phase 1 complete	Publisher, solutions, environment variables, connection references, control notes, and trial constraints validated.
Dataverse schema complete	Global choices, Lead/Opportunity/Account/Contact extensions, custom log/ledger tables, keys, duplicate detection.
Sales workspace complete	Forms, tabs, sections, views, roles, field-level security and seller experience validated.
Customer Insights plane ready	Compliance profile, domain, triggers, segments, emails, journeys, consent writeback and ledger configured.
SAP boundary ready	Azure resources, APIM, Functions, Key Vault, SAP wrapper, endpoints, idempotency, telemetry and API tests complete.
Sales AI agents ready	Sales Qualification and Opportunity/Deal Close agents configured in governed mode and tested.
Foundry ready	Parent orchestrator, child agents, canonical schemas, tool controls, safety policy, evaluation datasets validated.
Power Automate ready	Solution-aware flows, env variables, connection references, scopes, retries, logs, approvals tested.
Operations ready	Dashboards, alerts, SLAs, runbooks, support roles, replay process, incident response ready.
E2E tests passed without HubSpot	Lead, SAP context, opportunity risk, approval/SAP submit, disagreement safety tests passed.

Checkpoint	Required evidence
HubSpot last	Ingress mapping tested only after core ASO path is stable.

Approval gates

Gate	Approver	Approval question
Naming gate	Solution Architect + CRM Architect	Do component names and schema names follow the standard exactly?
Schema gate	CRM Architect + Security Architect	Are data types, security, keys, and duplicate detection approved?
Communication gate	Marketing Ops + Compliance	Are consent, compliance profile, preference center, and journey controls valid?
SAP gate	Integration Architect + Security Architect	Is all SAP access through APIM/Functions with idempotency and telemetry?
AI gate	AI Architect + Solution Architect	Do agents follow responsibility boundaries and evaluation standards?
Release gate	Platform Owner + Stream Leads	Are tests, runbooks, monitoring, rollback, and approvals complete?